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Clinical and Translational Gastroenterology Publish Ahead of Print

DOI: 10.14309/ctg.0000000000000933

Identification of biomarkers for treatment escalation of ulcerative colitis based on untargeted metabolomics and machine learning algorithms: A prospective cohort study

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CONFLICTS OF INTEREST

Guarantor of the article: Yongdong Wu, MD.

Specific author contributions: Muzhou Han, Haiyun Shi, and Yongdong Wu designed the study. Muzhou Han, Hao Wang, Siying Zhu, and other authors were involved in patient enrolment and performing the study. Hao Wang and Muzhou Han analysed the data and interpreted the results. Muzhou Han wrote the manuscript. Yongdong Wu was involved in the drafting of the manuscript and critical revision of the manuscript. All authors approved the final version of the manuscript.

Financial support: This work was funded by National Key R&D Program of China (2023YFC2507300); National Natural Science Foundation of China (No. 81600432); Grant No. PYZ23079 from Capital Medical University.

Potential competing interests: None to report.

IRB approval statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committees of Beijing Friendship Hospital Affiliated to Capital Medical University (No. 2017-P2-094-03).

ACKNOWLEDGMENTS

We acknowledge the investigators and patients who participated in this study.

Study Highlights

WHAT IS KNOWN

1. Biomarkers for guiding UC treatment escalation remain clinically unmet.
2. Untargeted metabolomics reveals metabolic shifts in inflammatory diseases.
3. Machine learning aids in biomarker-based disease prediction models.

WHAT IS NEW HERE

1. Nine plasma metabolites identified as UC treatment escalation biomarkers.
2. Phenylalanine/ether lipid metabolism pathways linked to treatment escalation.
3. Machine learning integrated metabolite biomarkers into UC treatment escalation prediction models.

Abstract

Background: Biomarkers to guide clinical decision making in active ulcerative colitis (UC) patients are urgently needed. This study aims to identify metabolites associated with UC treatment escalation and establish prediction models based on untargeted metabolomics and machine learning algorithms.

Methods: Liquid chromatography-mass spectrometry-based untargeted metabolomics analysis was performed on 88 plasma samples (44 active UC patients requiring treatment escalation and 44 active UC patients not requiring treatment escalation). Univariate and multivariate analyses were applied to identify metabolic biomarkers for UC treatment escalation. Metabolic pathway enrichment analysis was performed to reveal the disturbed metabolic pathways related to UC treatment escalation. Four machine learning algorithms, including Support Vector Machine (SVM), Random Forest, k-Nearest Neighbor (KNN), and logistic regression were used to build diagnostic models for UC treatment escalation.

Results: Nine significantly differential metabolites were identified as the candidate biomarkers for UC treatment escalation. Pathway analysis revealed that phenylalanine metabolism and ether lipid metabolism are the disturbed metabolic pathways related to treatment escalation. The protein-metabolite interaction network identified 21 proteins are associated with 9 treatment escalation related metabolites. The areas under the receiver operating characteristic curve of the SVM, Random Forest, KNN, and logistic regression models based on metabolic biomarkers were 0.923, 0.966, 0.897 and 0.803, respectively.

Conclusion: The plasma metabolome represents a promising source of biomarkers for the prediction of treatment escalation in active UC. Metabolic biomarkers, combined with machine learning algorithms, could be efficient for risk assessment and early identification of UC treatment escalation.

Keywords: Ulcerative colitis; Treatment escalation; Metabolomics; Machine learning algorithm.

1.Introduction

Ulcerative colitis (UC) is a chronic progressive inflammatory bowel disease (IBD) affecting the rectum and colon characterised by a relapsing and remitting course [1]. The clinical features of UC mainly include bloody diarrhoea, abdominal pain, fatigue and faecal incontinence [2], leading to a marked impairment in the quality of life of patients. The etiology of UC is still not fully understood. Multiple factors including an array of susceptibility gene variants, environmental factors and dysregulated immune response have been suggested to contribute to UC pathogenesis [3]. In recent decades, the incidence of UC has increased globally [4]. It is estimated that the prevalence of UC around the world is more than 5 million cases [5]. UC places a significant burden facing healthcare systems worldwide, which is likely to increase with time.

The principal aims of UC medical management are to induce and maintain remission, decrease the risk of complications, and improve quality of life [6]. Available pharmacological treatment is determined on the basis of the severity and extension of disease. 5-aminosalicylic acid (5-ASA) has always played a crucial first-line role in the therapy for UC [7-8]. However, some UC patients, especially a portion of moderate to severe UC, may fail to achieve remission on this therapy and, therefore, require treatment escalation, such as the use of corticosteroids and biologic therapies. Considering that early and effective treatment is the key to improving patient outcomes [9], it is crucial to identify patients who may require upgraded treatment in the future from all active UC patients.

In the last decade, in order to comprehensively characterize the metabolic profile of IBD and its role in the development of disease activity states, attempts have been made to screen for biomarkers using a variety of omics technologies, including metabolomics [10]. Metabolomics is a direct readout of the function of a system, which provides insight into the cellular processes in response to certain stimuli or interactions [11]. Compared to targeted metabolomics, untargeted metabolomics allows for the measurement of as many metabolites as possible from biological samples without a priori knowledge, with the advantage of discovering novel metabolites relevant to the study context.

Given the absence of prior research utilizing high-throughput data to explore the differences in plasma untargeted metabolomics among UC patients who have undergone different clinical strategies, we conducted this work to identify predictive biomarkers for UC treatment escalation.

In addition, we constructed the diagnostic models for identifying UC patients in need of therapeutic upgrades based on potential metabolic biomarkers using four machine learning algorithms, including Support Vector Machine (SVM), Random Forest, k-Nearest Neighbor (KNN), and logistic regression.

2. Materials and methods

2.1 Study design and participants

In this prospective cohort study, we consecutively recruited patients with active UC (Mayo score 3-12) from the Beijing Friendship Hospital Affiliated to Capital Medical University between August 2020 and September 2023. Patients who had received treatment escalation such as glucocorticoids, immunosuppressants, biologics, or surgery in the 6 months prior to enrollment were excluded. After obtaining written informed consent, we collected the peripheral blood samples and information on demographics, clinical characteristics, laboratory examination and colonoscopy results of the participants. Blood samples were tested using liquid chromatography-mass spectrometry (LC-MS) to quantify levels of metabolites. All patients received only 5-Aminosalicylate (5-ASA) as background therapy and were followed up for a minimum of 8 weeks after enrollment. During this period, experienced clinicians assessed the need for treatment escalation based on the Toronto Consensus [12] and made corresponding clinical decisions. The standard for treatment escalation is 5-ASA failure [12] during follow-up period. To provide clarity, 5-ASA failure was explicitly defined as the inability to achieve clinical remission despite optimal treatment with an adequate dose of oral or topical 5-ASA for a sufficient duration [12]. Clinical remission was defined as a Mayo score of ≤ 2 with no individual subscore exceeding 1. Consequently, treatment escalation was initiated when a patient failed to achieve clinical remission and required the addition of glucocorticosteroids, immunosuppressants, or biologic agents. Treatment escalation defined as the addition of glucocorticosteroids to achieve remission due to ineffective control of disease activity with existing 5-ASA regimens, or escalation to immunosuppressants or biologics. The classification criteria for the non treatment escalation group are defined as achieving clinical remission within 8 weeks using 5-ASA monotherapy alone, without requiring additional treatment escalation. Then, we divided all participants into the treatment escalation group and the non treatment escalation group based on their actual situation.

This study was approved by the Ethics Committees of Beijing Friendship Hospital Affiliated to Capital Medical University (No. 2017-P2-094-03). The study protocol adhered to the tenets of the Declaration of Helsinki. All participants provided signed informed consent.

2.2 Blood sample collection

Each peripheral blood sample was collected using a tube containing ethylenediaminetetraacetic acid (EDTA) from the study individuals on an empty stomach in the morning. After a 30-min standing period, the blood was centrifuged at 3000 rpm (1157g) at room temperature (22 °C) for 10 min. The upper plasma was collected and stored in freezers at – 80 °C until the metabolomics assay.

2.3 Metabolic profiling

The plasma samples were thawed at 4 °C. Once liquid, we obtained 100 µL sample and added 700 µL extract solution (methanol: acetonitrile: deionized water= 4: 2: 1, V/V/V), vortexed for 1 min. The whole samples were extracted by ultrasonication for 10 min and stored at – 20 °C for 2 h. The extract was centrifuged at 25,000 g at 4 °C for 15 min. A total of 600 µL of supernatant was transferred into new polypropylene tubes and dried in a freeze concentration centrifugal dryer. The powder was then resolubilized by adding 180 µL of 50% methanol in water, vortexed for 1 min, and extracted by ultrasonication for 10 min. After vigorous mixing, samples were re-centrifuged at 25,000 g at 4°C for 15 min. The supernatant was then removed for further analysis. To evaluate the stability and reproducibility of LC-MS analysis, a quality control (QC) sample was prepared by mixing each equal volume (20 µ L) of the supernatant to be analyzed.

LC-MS analysis was performed using an LC-MS system consisting of an ACQUITY ultra-performance liquid chromatography (UPLC) I-Class PLUS system (Waters Corporation, Milford, USA) coupled with a Q Exactive Focus Hybrid Quadrupole-Orbitrap mass spectrometry (Thermo Fisher Scientific, Waltham, USA). Chromatographic separations were accomplished by ACQUITY UPLC BEH C18 column (1.7 µm, 2.1×100 mm; Waters Corporation, Milford, USA) maintained at 45°C through the column oven. The mobile system consisted of ultrapure water with 0.1% (V/V) formic acid (mobile A) and methanol with 0.1% (V/V) formic acid (mobile B). The gradient was 0–1 min, 2% B; 1–9 min, 2–98% B; 9–12 min, 98% B; 12–12.1 min, 98–2% B; 12.1–15 min, 2% B. The constant flow rate was set to 0.35 mL/min and the sample injection volume was set to 5.0 µL.

The ion source was electrospray ionization (ESI), and the sample mass spectrometry signal acquisition was performed in positive and negative ion scanning mode respectively, with the scan range of 70–1050 mass-to-charge ratio (m/z). The source ion parameters applied were as follows: auxiliary gas heater temperature of 350 °C, sheath gas flow rate of 40 arbitrary units, auxiliary gas flow rate of 10 arbitrary units, capillary temperature of 320 °C, spray voltage of 3.80 kV (positive ion mode) and 3.20 kV (negative ion mode).

2.4 Data analysis and statistics

Raw mass spectrometry data obtained were transferred to the format of “mzXML” using ProteoWizard msConvert [13]. The XCMS platform (Version 4) was utilized to process raw chromatographic data, including feature detection, retention time correction and alignment [14]. The resultant feature matrix comprised information on mass-to-charge ratio (m/z), retention time and peak intensities.

Univariate and multivariate analyses were performed to delineate differences in metabolic profiles between two groups. The methods included parameter tests and nonparametric tests, differential expression multiple analysis, and orthogonal partial least square discriminant analysis (OPLS-DA). To screen for differential metabolites, variable influence on projection (VIP) values > 1 and $p < 0.05$ were selected. To control the false discovery rate (FDR) arising from multiple hypothesis testing, the Benjamini-Hochberg procedure was applied, and an FDR-adjusted p -value (q -value) threshold of < 0.05 was used to ensure that fewer than 5% of the identified significant metabolites were expected to be false positives. The metabolites and metabolic pathways were annotated by matching the exact molecular mass data (m/z) to those in the database within a 10-ppm threshold using the open-access databases including Kyoto Encyclopedia of Genes and Genomes (KEGG) [15] and Human Metabolome Database (HMDB) [16]. The correlation between individual metabolites was analyzed by calculating Spearman correlation coefficients between all metabolites. A system network of protein-metabolite interactions was created using the online processing tool of metabolomics data, MetaboAnalyst 4.0 ([http:// www.metaboanalyst.ca/](http://www.metaboanalyst.ca/)). Interaction networks were plotted using Cytoscape 3.7.1 (National Institute of General Medical Sciences, Bethesda, USA). The diagnostic models for treatment escalation were constructed by using 4 machine learning algorithms, including SVM (“e1071” R package), Random Forest (“randomForest” R package), KNN (“kknn” R package), and logistic regression (“glm” R package).

Receiver operating characteristic (ROC) curves were generated to quantify the predictive accuracy of the models, and the area under the ROC curve (AUC) was used to assess the discriminatory ability of the models.

All statistical analyses were performed using R version 4.1.2 (R Foundation for Statistical Computing) and SPSS 26.0 (SPSS Incorporation, Chicago, IL, USA). The Shapiro-Wilk test was used to test normality distribution of all variables. The student's t-test was used to compare two groups of continuous variables and to assess the statistical significance of normally distributed variables. The independent and the differences among non-normally distributed variables were analyzed using the Mann-Whitney U test. Chi-squared test or Fisher's exact test was used to compare and analyze statistical significance between two groups of categorical variables. A *p* value of 0.05 or less was considered statistical significance.

2.5 Machine learning modeling pipeline

To ensure the robustness and generalizability of our predictive models, a comprehensive machine learning pipeline was rigorously implemented. The peak intensities of the nine identified metabolic biomarkers served as the input features. The data preprocessing involved imputing missing values using the k-nearest neighbor ($k=5$) algorithm and standardizing all features to a z-score distribution to mitigate scaling biases. The entire cohort of 88 samples was randomly partitioned into a training set (70%) for model development and a hold-out test set (30%) for final performance evaluation. Model training and hyperparameter tuning were conducted on the training set employing a 5-fold cross-validation strategy to optimize model parameters and guard against overfitting. Four distinct algorithms—SVM, Random Forest, KNN, and logistic regression—were constructed and evaluated. The final performance of each model was assessed on the independent test set, with key metrics including accuracy, sensitivity, specificity, and AUC being reported. Furthermore, to enhance interpretability, feature importance for the optimal model was analyzed using the Gini importance metric from the Random Forest algorithm. The logistic regression model was implemented as a binomial classifier using the glm function in R with the 'binomial' family. The model was trained to predict the binary outcome of treatment escalation (coded as 1 for the escalation group and 0 for the non-escalation group). The predictor variables (fixed effects) in the model were the standardized peak intensities of the nine significant metabolic biomarkers identified from our differential analysis. Given the structure of our data from a

single-center cohort without inherent hierarchical clustering, no random effects were specified in the model configuration. Model performance was evaluated on the independent test set using the same metrics as the other machine learning algorithms. This structured approach was designed to ensure methodological rigor and the reliability of our findings.

3. Results

3.1 Clinical characteristics of the study population

In total, 44 treatment escalation UC patients and 44 non treatment escalation UC controls were analysed in the present study. The details about the demographic, biochemical, colonoscopic, and anthropometric characteristics of the treatment escalation UC patients and non treatment escalation UC controls are presented in Table 1. The average ages of the treatment escalation group and control groups were 47.09 ± 17.11 and 49.68 ± 15.66 years, respectively. The extension were more extensive in treatment escalation group patients ($P < 0.05$). The Mayo endoscopic score and Ulcerative Colitis Endoscopic Index of Severity in treatment escalation group were significantly higher compared to those in control group ($P < 0.05$). Erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) levels were also higher in the treatment escalation group ($P < 0.05$). Aside from these, there were no statistically significant differences observed between the two groups in terms of the other variables ($P > 0.05$).

3.2 Identification of potential metabolic biomarkers

LC-MS technique was used to perform metabolomics analysis on 88 plasma samples and 8 QC samples. The total ion chromatograms of the QC samples are presented in Supplementary Figure 1. Unsupervised model principal component analysis (PCA) of all samples showed tight aggregation of QC samples (Supplementary Figure 2), indicating the robustness of the metabolic assay platform operation. In QC samples, the number of metabolites with coefficient of variation (CV) less than 30% accounted for 70% of the total metabolites, supporting the robust quality of our data and the reproducibility of sample preparation method.

After feature detection, retention time correction, alignment, and removal of missing values, 3317 metabolites were identified successfully. The OPLS-DA model showed that the treatment escalation group was separated from the no-need treatment escalation group, indicating the levels of metabolites in treatment escalation participants were different from the controls (Figure 1A). Permutation tests were used to prevent overfitting of the established model. The OPLS-DA

model yielded a cumulative R^2Y of 0.981 and a cumulative Q^2 of 0.82, which indicates that the model has excellent goodness of fit, reliability and predictive ability. Among 3317 detected features, 360 detected features with VIP values > 1 , P values < 0.05 , FDR adjusted P values < 0.05 , and fold change > 1.2 (or < 0.83) were selected as the candidate features for UC treatment escalation (Figure 1B and Supplementary Table 1). Among them, 265 metabolites were detected in positive ion mode (24 compounds showed upward trends while 241 compounds showed downward trends), and 95 metabolites were detected in negative ion mode (14 compounds showed upward trends while 81 compounds showed downward trends). After adjusting for the confounders, 9 of the 360 differential features were identified successfully as potential biomarkers (Table 2). Of these, levels of 8 metabolites were decreased, including 12-hydroxydodecanoic acid, canthaxanthin, phenylacetaldehyde, 3,5-dihydroxybenzoic acid, benzene, amantadine, azelaic acid, and theophylline. Conversely, level of 4-aminophenol was significantly increased in UC treatment escalation group.

3.3 Metabolic pathway enrichment analysis

Quantitative enrichment analysis showed that the top 10 enriched metabolite sets identified by the difference in the metabolic profiles between the treatment escalation UC group and the non-treatment escalation UC group were phosphatidylcholine catabolism, G protein-coupled receptor (GPCR) ligand binding, classical antibody-mediated complement activation, signaling by GPCR, fatty acid omega-oxidation, purinergic signaling, peptide hormone metabolism, class A/1 (Rhodopsin-like receptors), the Nod-like receptor family pyrin domain containing 3 (NLRP3) inflammasome, and hydrolysis of lysophosphatidylcholine (LPC) (Figure 2A). The pathways with a P value less than 0.05 were considered as significantly enriched pathways, and all the top 10 enriched metabolite sets mentioned above are significantly enriched pathways.

In order to reveal the disturbed metabolic pathways related to UC treatment escalation, the pathway analysis of potential metabolic biomarkers was performed using Metaboanalyst and information from KEGG and HMDB database. These metabolites were involved in 5 metabolic pathways. Among these 5 metabolic pathways, two pathways with P values < 0.05 and impact values > 0.00 were identified as phenylalanine metabolism and ether lipid metabolism, respectively (Figure 2B). Of these, phenylalanine metabolism had the most significance and the largest impact.

3.4 Protein-metabolite interaction network

The protein-metabolite interaction network is a visualisation and exploration of the interactions between metabolites and functionally related proteins. Using HMDB and Search Tool for Interactions of Chemicals (STITCH) database [17], we constructed a protein-metabolite interaction network to provide the comprehensive understanding of potential functional relationships among differently expressed proteins and potential metabolic biomarkers. Network analysis revealed that 21 proteins interacted directly with 9 metabolic biomarkers for UC treatment escalation (Figure 3).

3.5 Diagnostic models for treatment escalation using machine learning algorithms

After comprehensively profiling the metabolic biomarkers, four machine learning algorithms, including SVM, Random Forest, KNN, and logistic regression, were performed to construct diagnostic models based on 9 metabolic biomarkers. We randomly split all samples into 70% training dataset and 30% testing dataset to assess the overall performance of every model, and subsequently performed the ROC curve analysis on the training and test datasets. The diagnostic performance of these four models was shown in Table 3 and Figure 4. The accuracy of models in both training set and testing set was greater than 81.8 % in distinguishing treatment escalation UC group and no-need treatment escalation UC group, and the AUCs ranged from 0.803 to 0.966. Notably, we attempted to incorporate conventional clinical indices, such as haemoglobin, platelets, CRP, albumin, and ESR, into the models but did not observe significant improvements (data not shown).

3. Discussion

The need for UC treatment escalation is the result of clinical judgement in light of all the investigations and scores available to the treating physician and the tolerability of symptoms for each individual patient [18]. Currently, neither the characteristics of the colonoscopy nor a number of common non-invasive markers can be considered as independent predictors of disease clinical progression [19]. There is an urgent need for biomarkers to guide clinical decision making in active UC, and personalized approaches based on biomarkers remain one of the most attractive topics in UC. Metabolomics is a rapidly developing technology that is being applied in many areas, including biomarker investigation. However, the metabolomic characteristics of treatment escalation UC patients have not been adequately studied and are currently poorly understood. As

far as we are aware, this prospective cohort study is the first to provide a comprehensive set of metabolic biosignatures of active UC patients in need of treatment escalation during the subsequent course of the disease using LC-MS-based metabolic profiling. Metabolic biosignatures revealed differences between active UC patients requiring escalated therapy and those not requiring escalated therapy. This provides a rational basis for personalised therapy in IBD and represents an important step towards precision medicine for patients with UC.

In recent years, the metabolic approach has been increasingly used in a number of studies of experimental and human IBD, where the focus has been on the investigation of disease-related metabolites in order to gain more insight into metabolic pathways. Recent studies have identified several metabolic pathways associated with the severity of UC disease activity, including amino acid metabolism, phospholipid metabolism, glutathione production and tricarboxylic acid cycle [20-24]. Some metabolites involved play roles in regulating intestinal inflammation and intestinal mucosal homeostasis, which may be related to energy metabolism, intestinal barrier integrity, regulation of oxidative stress and inflammation, and gut microbiota [25-28]. Complex analysis of biological samples at the untargeted metabolomics level to quantify and characterise relevant metabolites also represents a promising strategy for the discovery of potential therapeutic targets [29]. In our study, based on the 9 potential metabolic biomarkers for active UC treatment escalation, the pathway analysis suggested that phenylalanine metabolism, and ether lipid metabolism pathways were disturbed in UC patients requiring intensified treatment.

Phenylalanine is one of the essential amino acids belonging to the aromatic amino acids. Most of phenylalanine is oxidised in the body to form tyrosine, which is involved in the synthesis of important neurotransmitters and hormones such as dopamine and catecholamine hormones [30]. Currently, many studies have shown metabolic differences in plasma of IBD patients, with amino acid changes depending on colitis severity [31-32]. Among these pathways, phenylalanine metabolism is the most affected [33]. New evidence [34] discovered that UC patients whose endoscopic activity worsened (Ulcerative Colitis Endoscopic Index of Severity increase ≥ 1) over 6 months had lower baseline plasma phenylalanine and tyrosine concentrations levels compared with UC patients whose endoscopic activity improved (Ulcerative Colitis Endoscopic Index of Severity decrease ≥ 1). Previous studies have also found that IBD patients have higher faecal phenylalanine levels compared to healthy controls [35-37], which may in part be due to

inflammation-related malabsorption in these patients. In our study, the different plasma metabolite profile related to phenylalanine metabolism could discriminate between active UC patients requiring treatment escalation and those do not. For example, UC treatment escalation patients have significantly lower levels of phenylalanylphenylalanine, phenylalanyltryptophan, phenylalanylleucine, and phenylalanyl-phenylalanyl-arginine chloromethyl ketone. This indicated that disturbances in phenylalanine metabolism may be associated with the pathophysiology of UC and may serve as a potential predictive biomarker for the progression of disease.

Ether lipids are a group of glycerolipids in which the hydrocarbon chain is attached to an ether linkage at the sn-1 position of the glycerol backbone [38-39]. In mammals, over 20% of all glycerophospholipids are classified as ether lipids [40]. Interest in ether phospholipids has increased with the advances in MS analysis over the past few years. As the main structural components of cellular membranes, ether lipids were found mainly within the lipid classes of phosphatidylcholine (PC) and phosphatidylethanolamines (PE) [41], which are thought to be related to intestinal metabolic balance. The intestine of UC patients is a complex pathological environment, including oxidative stress, osmotic stress, DNA damage, and abnormal immune response [42-43]. The current study shows that ether lipids are involved in shaping membrane properties, act as potent antioxidants and are involved in inflammatory signalling [44]. Altered ether lipid levels have been observed in several diseases, such as rhizomelic chondrodysplasia punctata, peroxisomal dysfunction, neurodegenerative diseases, cardiovascular diseases, cancer, and metabolic disorders [44-45]. In our study, the results of pathway analysis suggest that ether lipid metabolism disorders may be associated with the need for treatment escalation in active UC patients. Active UC patients necessitating treatment escalation exhibit significantly lower plasma baseline concentrations of the ether lipids 1-O-(2R-hydroxy-4Z-eicosenyl)-sn-glycol and 1-O-(2R-hydroxy-4Z-nonadecenyl)-sn-glycol, suggesting their potential roles in the inflammatory response of IBD. The 4th position double bond and 2R-hydroxy may confer antioxidant or signalling properties to the molecules, but this possibility requires further investigation.

Notably, in our research, the major enriched metabolite sets identified by the difference in metabolic profiles between two groups include PC catabolism, G protein-coupled receptor (GPCR) ligand binding, signaling by GPCR, and hydrolysis of LPC. Recent research has shown that PC has a promising potential in the treatment of IBD. Van et al. [46] indicated that PC are important

protective components of colonic mucus secreted by intestinal cells. Supplementing the colon mucosa of UC patients with a PC-rich phospholipid mixture has shown significant therapeutic efficacy in UC [47-48]. Yu et al. [49] found that exogenous PC 34:1 could exert anti-UC effects by reversing the reduction of fumarate in mouse colon. PC could also release lysophosphatidylcholine (LPC) and esterified unsaturated fatty acids to form bioactive pro-inflammatory and anti-inflammatory mediators when metabolised by phospholipase A₂ [50]. LPC is further hydrolyzed by lysophospholipases to generate glycerophosphocholine (GPC) and free fatty acids. Current evidence indicates that LPC, through GPCR signaling, has deleterious effects on various cells, including enhancing inflammatory responses, disrupting mitochondrial integrity, and inducing apoptosis [51]. Pseudo-targeted lipidomics also revealed accumulation of choline-containing compounds in the gut during UC exacerbation [52]. Similarly to previous research findings, in our study, it was observed that the levels of PC (14:0/14:0), PC (16:1 (9Z)/14:1 (9Z)), PC(18:1(6Z)/0:0), and phosphatidic acid (19:0/12:0) were significantly lower in UC patients who require treatment escalation, while the level of GPC was significantly higher. These findings indicate the presence of excessive PC and LPC hydrolysis in these patients, suggesting their important role in the clinical progression of UC.

Necessity and timing of treatment escalation in active UC have been used as robust endpoints in some current clinical trials [53-54]. Compared to other study endpoints such as mucosal healing, medication changes are not always immediately pre-empted by colonoscopy and are also more clinically significant. In our study, the need for treatment escalation in the subsequent course of active UC patients was actually closely related to 5-ASA failure, which is inability of the patient to achieve and maintain complete corticosteroid-free remission despite optimal treatment with oral, rectal, or combination 5-ASA therapy [12]. Given that 5-ASA failure may be one of the important intrinsic reasons for escalating treatment, its association with related metabolic disorders and the specific mechanism of action also warrant further investigation.

While our initial untargeted analysis identified a broad list of 360 potential biomarkers, we employed a multi-step statistical refinement process to mitigate the risk of false positives and ensure the robustness of subsequent modeling. The final panel of nine metabolic biomarkers used for machine learning was derived after a more stringent adjustment for potential confounders. Conventional statistics-based models are usually not suitable for the analysis of

metabolomic big data because of its complex structure, including high dimensionality, high noise and missing values [55]. Machine learning algorithms are therefore becoming increasingly important in the data mining of metabolomics, especially in building models on the basis of potential biomarkers for diagnosing diseases and predicting progression [56]. For example, Su and colleagues [57] used an algorithm based on ten metabolites to predict the future development of type 2 diabetes in Chinese people. Wang et al. [58] constructed diagnostic models for metabolic syndrome using four machine learning algorithms based on metabolic biomarkers. Notably, it is also crucial to select and optimise the machine learning algorithms. In our study, four machine learning algorithms, including SVM, Random Forest, KNN, and logistic regression were used to build identification models for UC treatment escalation. ROC curve analysis showed that the AUCs of four models based on metabolic biomarkers ranged from 0.803 to 0.999, underscoring their potential clinical utility and relevance in precision medicine for UC patients.

There are several limitations of this work. Firstly, the number of patients in the study was small, and the study was conducted only in a single Chinese institution. To generalise findings to a broader spectrum of population, validation needs to be done in the larger multi-centre cohorts. Secondly, it is difficult to determine whether the observed UC treatment escalation related metabolite levels disruption are the cause or the phenotype. Follow-up Mendelian randomisation studies are needed for causal inference, as well as basic research on the corresponding metabolites to elucidate whether they play a role in the the action mechanism. Thirdly, in view of the semi-quantitative nature of the untargeted metabolomics approach, a targeted metabolomics study is being conducted on the same cohort to validate the putative biomarkers and pathways on the basis of the findings in this study. Furthermore, regarding the machine learning analysis, the relatively small sample size relative to the number of features raises concerns about potential model overfitting, despite the application of cross-validation techniques. Although the models demonstrated promising performance on the internal test set, this risk implies that the reported predictive accuracy might be optimistic and may not generalize perfectly to external populations. Therefore, external validation in independent, larger cohorts is equally crucial to confirm the robustness and generalizability of the metabolite-based predictive models developed in this study. Despite these limitations, our findings provide preliminary evidence that plasma metabolomics could serve as an evidence-based and complementary tool to identify patients with active UC who

may require treatment escalation during disease progression, thereby supporting more informed clinical decision-making.

4. Conclusion

In this prospective cohort study, we identified 9 metabolic biomarkers (12-hydroxydodecanoic acid, canthaxanthin, phenylacetaldehyde, 3,5-dihydroxybenzoic acid, benzene, amantadine, azelaic acid, theophylline, 4-aminophenol) for predicting treatment escalation in active UC patients. Pathway analysis revealed that phenylalanine metabolism and ether lipid metabolism are the disturbed metabolic pathways related to treatment escalation. The protein-metabolite interaction network provided innovative insights into the pathophysiology of UC. These findings suggest that plasma metabolome is a promising biomarker resource for predicting UC treatment escalation. Metabolic biomarkers, combined with machine learning algorithms, could be efficient for risk assessment and early identification of UC treatment escalation.

Supplementary Figure 1-<http://links.lww.com/CTG/B406>

Supplementary Figure 2-<http://links.lww.com/CTG/B407>

Supplementary Table 1-<http://links.lww.com/CTG/B408>

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Table 1. Clinical characteristics of the UC patients

	Case (n = 44)	Control (n = 44)	P-value
Male	22 (50.00)	22 (50.00)	1.000
Age, years	47.09 ± 17.11	49.68 ± 15.66	0.461
BMI, kg/m²	20.41 ± 1.49	20.62 ± 1.51	0.511
Montreal classification			0.003
E1	1 (2.27)	11 (25.00)	
E2	10 (22.73)	13 (29.55)	
E3	33 (75.00)	20 (45.45)	
Mayo endoscopic score			<0.001
1	1 (2.27)	11 (25.00)	
2	18 (40.91)	29 (65.91)	
3	25 (56.82)	4 (9.09)	
Ulcerative Colitis Endoscopic Index of Severity	5.00 (4.00, 6.00)	4.00 (2.00, 4.25)	<0.001
Smoking	6 (13.64)	5 (11.36)	0.747
Drinking	8 (18.18)	7 (15.91)	0.777
CRP, mg/L	18.415 (0.30, 137.24)	5.93 (0.08, 67.73)	<0.001
ESR, mm/1h	24 (1, 102)	10.5 (2, 73)	<0.001

Note: Data are presented as means ± SD, medians (interquartile ranges), or frequencies (percentages). *P* < 0.05 was considered statistically significant. SD, standard deviation. CRP, C-reactive protein. ESR, erythrocyte sedimentation rate.

Table 2. Differential metabolites identified between active UC patients requiring treatment escalation and active UC patients who did not.

Metabolites	Case Group		Control Group		Trend	FC	VIP	P Value	P* value	P# value
	Mean	SD	Mean	SD						
4-Aminophenol	1.018	1.191	0.484	0.599	Up	2.103	1.239	0.006	0.017	0.020
12-Hydroxydodecanoic acid	0.352	0.344	0.781	0.752	Down	0.450	1.448	<0.001	<0.001	0.015
Canthaxanthin	7.206	9.935	16.606	9.788	Down	0.434	1.743	<0.001	<0.001	<0.001
Phenylacetaldehyde	0.128	0.165	0.302	0.312	Down	0.423	1.408	<0.001	0.001	0.005
3,5-Dihydroxybenzoic acid	3.657	6.141	9.676	9.458	Down	0.378	1.597	<0.001	<0.001	0.003
Benzene	0.13	0.296	0.638	1.146	Down	0.204	1.309	0.001	0.003	0.032
Amantadine	0.014	0.047	0.075	0.135	Down	0.184	1.247	<0.001	0.002	0.024
Azelaic acid	0.756	4.65	6.349	12.568	Down	0.119	1.197	<0.001	<0.001	0.027
Theophylline	0.001	0.004	0.011	0.021	Down	0.095	1.349	0.005	0.015	0.048

Note: FC, fold change; VIP, variable importance on projection. Down trend, relatively lower levels of metabolites present in MetS group. Up trend, relatively higher levels of metabolites present in the MetS group. *P* value, *P* value from Wilcoxon test without adjustment; *P** value, *P* value adjusted for false discovery rate using the *Benjamini–Hochberg* method. *P*# Value, *P* value adjusted for age and sex. *P* < 0.05 was considered statistically significant.

Table 3. Diagnostic performance of four machine learning algorithms

Machine learning algorithms	Models	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)	Accuracy (95% CI)	AUC (95% CI)
Support vector machine	Train	0.903 (0.731-0.980)	0.800 (0.563-0.943)	0.875 (0.691-0.987)	0.842 (0.603-0.968)	0.848 (0.698-0.942)	0.909 (0.827-0.967)
	Test	0.846 (0.546-0.981)	1.000 (0.663-1.000)	1.000 (0.692-1.000)	0.875 (0.553-0.997)	0.909 (0.696-0.990)	0.923 (0.786-1.000)
Random forest	Train	1.000 (0.891-1.000)	0.971 (0.846-0.999)	0.971 (0.846-0.999)	1.000 (0.891-1.000)	0.985 (0.923-1.000)	0.999 (0.995-1.000)
	Test	0.846 (0.546-0.981)	1.000 (0.663-1.000)	1.000 (0.692-1.000)	0.875 (0.553-0.997)	0.909 (0.696-0.990)	0.966 (0.872-1.000)
K-nearest neighbor	Train	0.839 (0.662-0.945)	0.886 (0.695-0.975)	0.867 (0.683-0.966)	0.861 (0.667-0.966)	0.864 (0.714-0.951)	0.918 (0.845-0.969)
	Test	0.769 (0.462-0.950)	0.889 (0.518-0.997)	0.909 (0.587-0.998)	0.727 (0.349-0.945)	0.818 (0.572-0.952)	0.897 (0.744-1.000)
Logistic regression	Train	0.742 (0.559-0.877)	0.971 (0.846-0.999)	0.958 (0.791-0.999)	0.810 (0.628-0.927)	0.864 (0.714-0.951)	0.900 (0.817-0.966)
	Test	0.846 (0.546-0.981)	0.889 (0.518-0.997)	0.917 (0.615-0.998)	0.800 (0.444-0.975)	0.864 (0.654-0.970)	0.803 (0.573-1.000)

Note: PPV, positive predictive value; NPV, negative predictive value; AUC, area under curve; CI, confidence interval; The 95% CIs of AUCs were estimated using bootstrap resampling for 2,000 times.

Figure legends

Figure 1. Metabolic biomarkers for UC treatment escalation.

(A) The orthogonal projection to latent structure-discriminant analysis (OPLS-DA) score plots. (B) Volcano plot of candidate metabolic biomarkers.

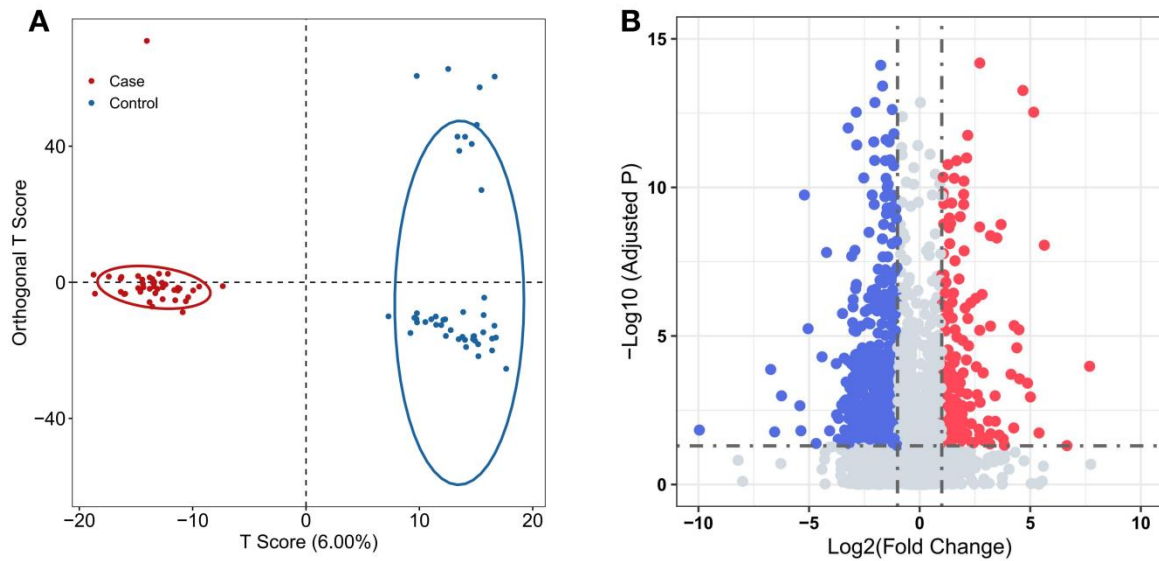


Figure 2. Disturbed pathways related to UC treatment escalation.

(A) Kyoto Encyclopedia of Genes and Genomes pathway enrichment analysis of UC treatment escalation. (B) Disturbed metabolic pathways in active UC patients necessitating treatment escalation.

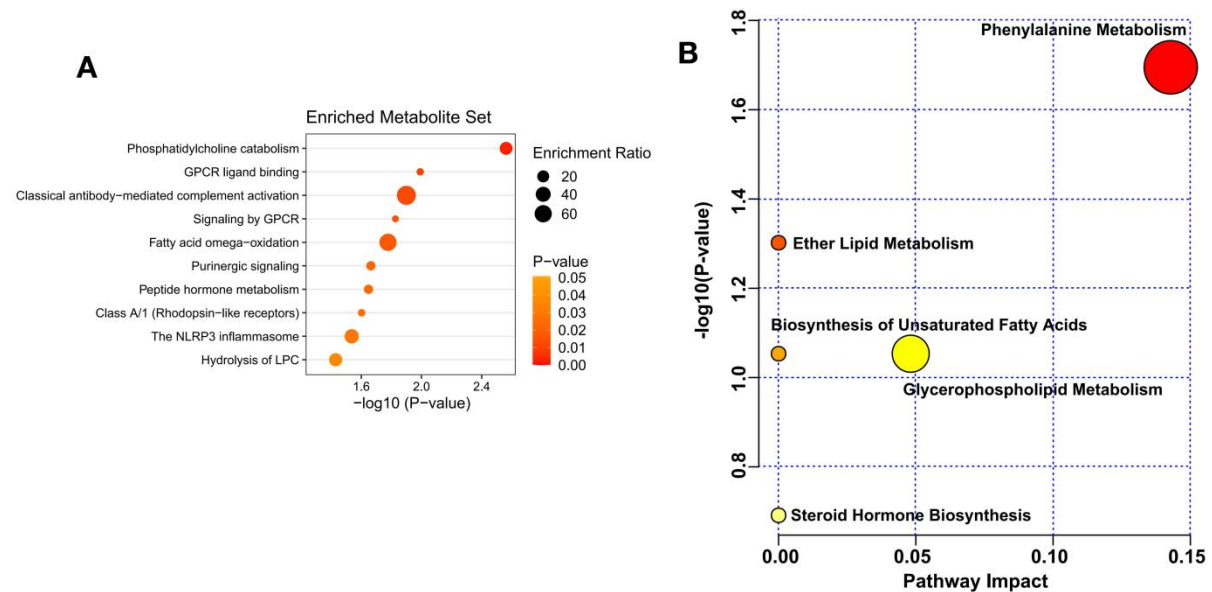


Figure 3. The metabolite-gene network analysis of potential metabolites for UC treatment escalation.

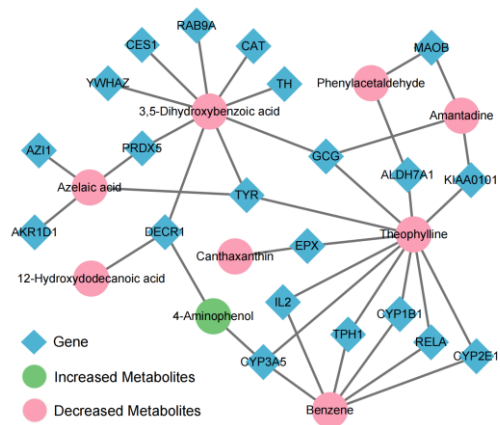
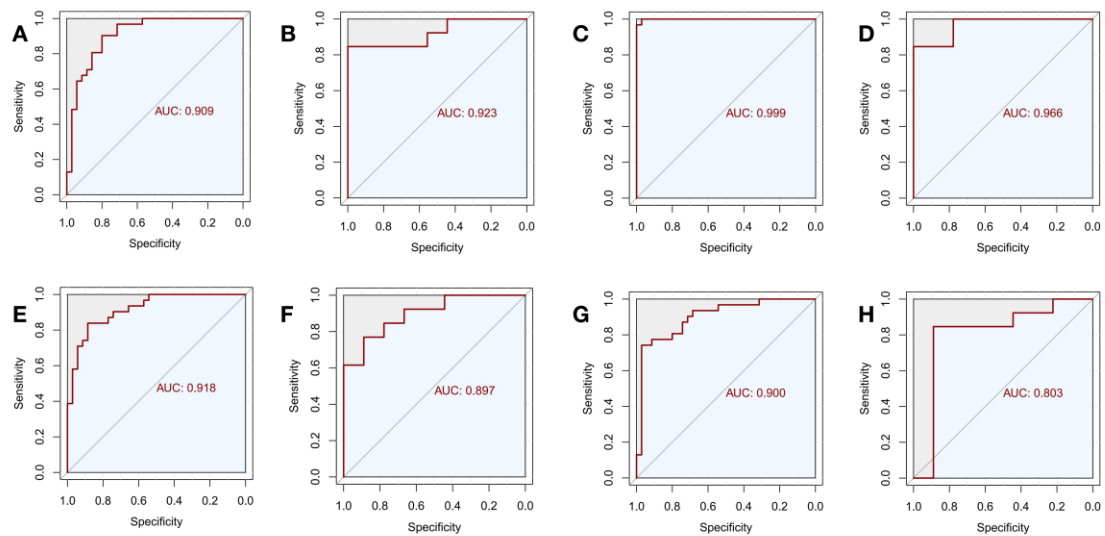


Figure 4. Area under the receiver operating characteristic curves of four machine learning algorithms based on metabolic biomarkers.

(A) Support vector machine model (train set); (B) Support vector machine model (test set) ; (C) Random forest model (train set); (D) Random forest model (test set); (E) K-nearest neighbor model (train set); (F) K-nearest neighbor model (test set); (G) Logistic regression model (train set); (H) Logistic regression model (test set).



Supplementary Figure 1. The total ion chromatograms of quality control plasma samples

(A) The total ion chromatograms of 8 quality control samples in positive ion mode; (B) The total ion chromatograms of 8 quality control samples in negative ion mode.

Supplementary Figure 2. The principal component analysis (PCA) score plots of all samples.

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